
NEXGENBIOLAB

Drug discovery projects

Protein preparation

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TASK | Protein Preparation

Objective: To download and prepare the protein structure for molecular docking using UCSF Chimera.

Method Summary:



Protein preparation is a critical pre-processing step in structure-based drug design. The raw PDB file downloaded from the Protein Data Bank typically contains water molecules, multiple chains, co-crystallized ligands, and missing atoms that must be addressed before docking. UCSF Chimera's Dock Prep tool automates the addition of hydrogens, assignment of Gasteiger charges, and removal of unwanted molecules, producing a clean, computation-ready structure.

► Task Steps

1	Go to RCSB PDB (https://www.rcsb.org) . Review the entry details (resolution, method, chains).
2	Download the PDB file (Download Files → PDB Format).
3	Open in UCSF Chimera (File → Open).
4	Inspect the structure: note the number of chains, water molecules, ligands, and cofactors present.
5	Select Chain A only (if other chains are present). Delete all other chains.
6	Remove all water molecules (Select → Residue → HOH → Actions → Delete).
7	Remove any unwanted cofactors or heteroatoms not needed for docking.
8	Run Dock Prep (Tools → Structure Editing → Dock Prep).
9	Save the prepared structure as prepared.pdb.
10	Open both the raw and prepared files in PyMOL. Take before-and-after screenshots.

Protein Structure Information

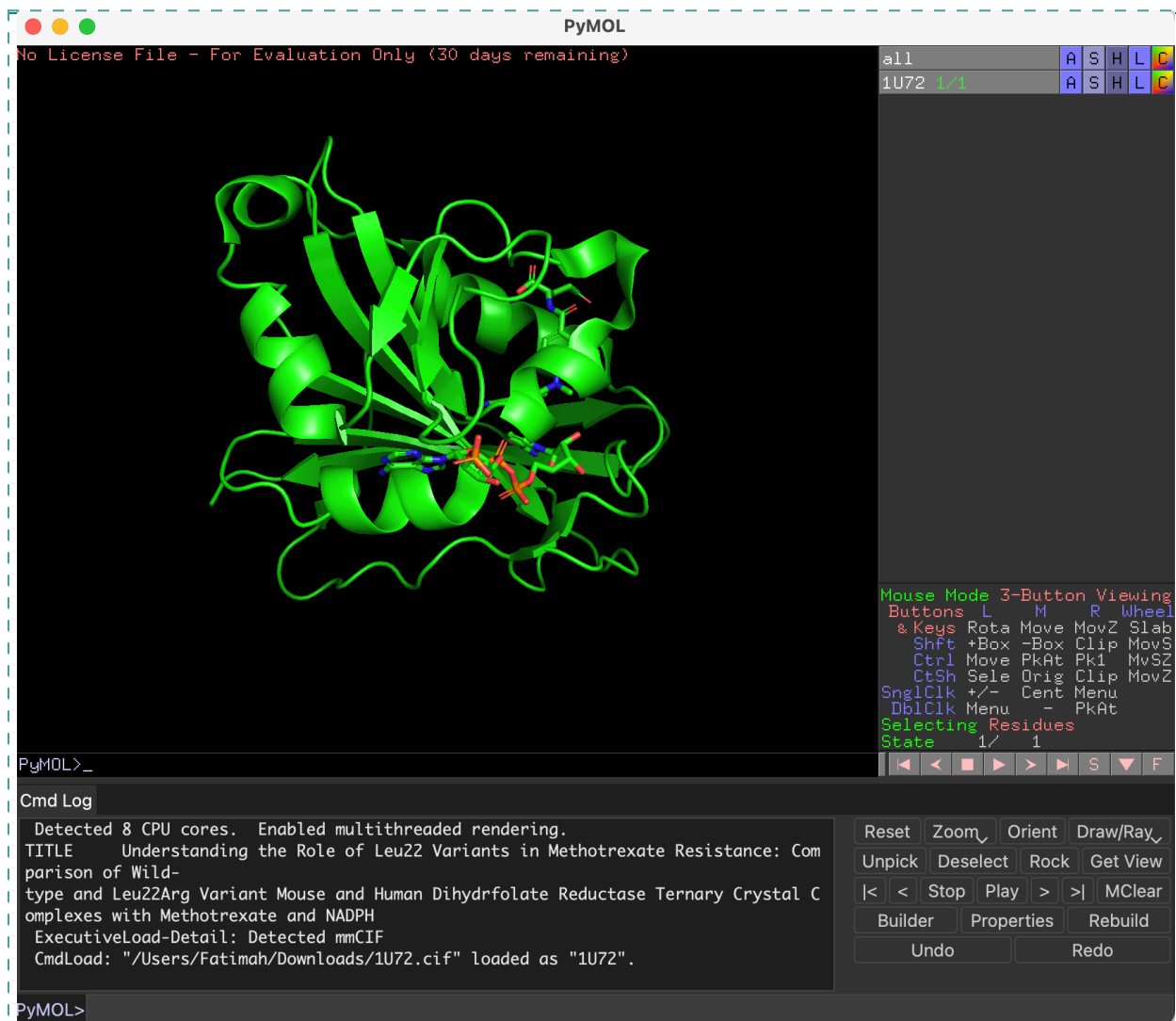
Property	Details
PDB ID	1U72
Protein Name	Dihydrofolate reductase (DHFR)
Experimental Method	X-ray diffraction
Resolution (Å)	1.90 Å

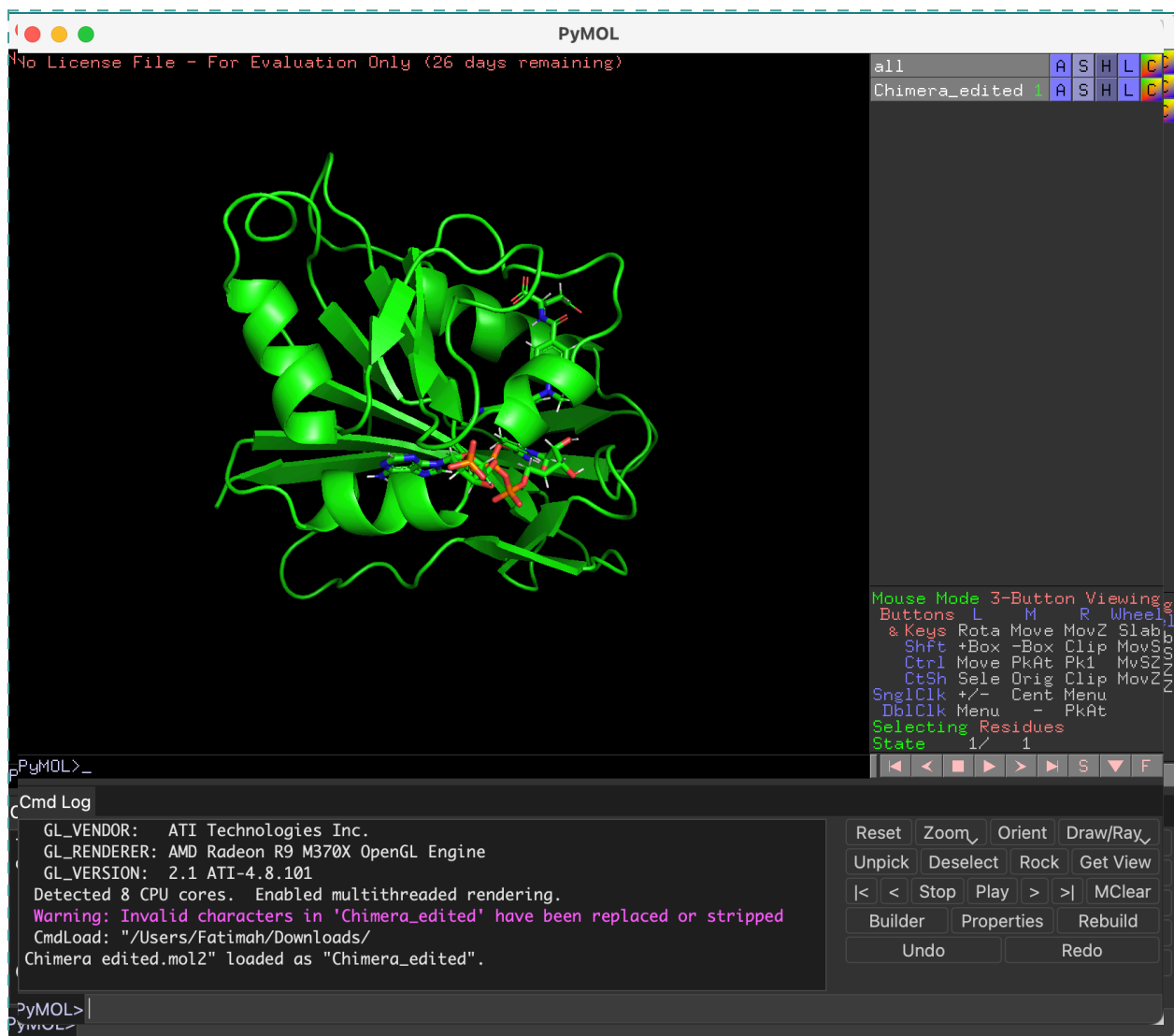
Co-crystallized Ligand(s)	NADPH and Methotrexate (MTX)
Water Molecules Removed?	<u>Yes</u> / No
Hydrogens Added via Dockprep?	<u>Yes</u> / No

Step-by-Step Preparation Log

Step	Action Performed	Observation / Result
1	Searched for PDB entry 1U72 in RCSB PDB.	The experimental structure of DHFR was identified.
2	Reviewed the PDB entry details, including experimental method, resolution, and chains.	The structure was determined by X-ray diffraction at 1.90 Å resolution
3	Downloaded the structure in PDB format from RCSB PDB.	The raw 1U72.pdb file was obtained.
4	Opened 1U72.pdb in UCSF Chimera.	The protein structure and associated molecules were displayed.
5	Inspected the structure and selected Chain A for preparation.	Chain A was retained as the protein structure for docking preparation.
6	Removed all water (HOH) molecules.	Water molecules were removed from the structure.
7	No unwanted cofactors or heteroatoms were identified, so no additional components were deleted.	Inspected the remaining ligands and heteroatoms to determine whether any additional components needed to be removed.
8	Ran Dock Prep from Tools → Structure Editing → Dock Prep.	Hydrogens were added and the structure was prepared for docking.
9	Saved the prepared structure as prepared.pdb.	A clean, prepared protein structure was successfully generated.
10	Opened the raw and prepared structures in PyMOL for comparison.	The prepared structure showed the removal of water/unwanted components and the addition of hydrogens compared with the raw structure.

Before & After PyMOL Screenshots





► Written Justification

?Why is protein preparation necessary before molecular docking

Protein preparation is necessary before molecular docking to ensure that the protein structure is clean and suitable for computational analysis. It removes unwanted molecules such as water and irrelevant heteroatoms, adds missing hydrogens, and assigns appropriate charges. These steps help provide a more accurate representation of the protein and improve the reliability of protein–ligand docking results.

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